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Work vehicle and control method for work vehicle

Abstract

A work vehicle includes a power source, a travel device that includes a clutch capable of adjusting a degree of engagement between the power source and a transmission, and travels on a basis of power transmitted from the power source, working equipment, a drive device that operates the working equipment on a basis of power transmitted from the power source, and a control device that outputs a control command for controlling the degree of engagement of the clutch on a basis of a state of the working equipment.

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Background/Summary

FIELD

(1) The present disclosure relates to a work vehicle and a control method for a work vehicle.

BACKGROUND

(2) In a technical field related to a work vehicle, a wheel loader that automatically excavates a target as disclosed in Patent Literature 1 is known.

CITATION LIST

Patent Literature

(3) Patent Literature 1: WO 2015/004809 A

SUMMARY

Technical Problem

(4) When excavating the target, the wheel loader moves forward toward the target and inserts at least a part of working equipment into the target. For example, in a case where the working equipment is strongly pressed against the target by the forward movement of the wheel loader, a situation in which the wheel loader cannot move may occur. When the situation in which the wheel loader cannot move occurs, work efficiency of the wheel loader is lowered.

(5) An object of the present disclosure is for a work vehicle to get out of a situation of being unable to move in excavation of a target.

Solution to Problem

(6) According to an aspect of the present invention, a work vehicle comprises: a power source; a travel device that includes a clutch capable of adjusting a degree of engagement between the power source and a transmission, and travels on a basis of power transmitted from the power source; working equipment; a drive device that operates the working equipment on a basis of power transmitted from the power source; and a control device that outputs a control command for controlling the degree of engagement of the clutch on a basis of a state of the working equipment.

Advantageous Effects of Invention

(7) According to the present disclosure, a work vehicle can get out of a situation of being unable to move in excavation of a target.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a side view illustrating a work vehicle according to an embodiment.
- (2) FIG. 2 is a diagram illustrating working equipment and a drive device according to the embodiment.
- (3) FIG. 3 is a configuration diagram illustrating the work vehicle according to the embodiment.
- (4) FIG. 4 is a functional block diagram illustrating a control system of the work vehicle according to the embodiment.
- (5) FIG. 5 is a block diagram illustrating a control device of the work vehicle according to the embodiment.
- (6) FIG. 6(A) and FIG. 6(B) are views for explaining an excavation operation of the work vehicle according to the embodiment.
- (7) FIG. 7(A) to FIG. 7(H) are timing charts illustrating a control method for the work vehicle according to the embodiment.
- (8) FIG. 8 is a flowchart illustrating the control method for the work vehicle according to the embodiment.
- (9) FIG. 9 is a flowchart illustrating the control method for the work vehicle according to the embodiment.
- (10) FIG. 10 is a flowchart illustrating the control method for the work vehicle according to the embodiment.
- (11) FIG. 11 is a diagram for describing an auto-tilt cycle table according to the embodiment.

DESCRIPTION OF EMBODIMENTS

(12) Embodiments of the present disclosure will be described below with reference to the drawings, and the present disclosure is not limited to these embodiments. Components of each of the embodiments described below can arbitrarily be combined. Also, some components may not be used.

(13) In the embodiments, a local coordinate system is set in a work vehicle **1**, and the positional relationship of the respective parts will be described with reference to the local coordinate system. In the local coordinate system, a direction extending in the right-left direction (vehicle width direction) as viewed from an operator on the work vehicle **1** is defined as a direction of an X axis, a direction extending in the front-rear direction of the work vehicle **1** is defined as a direction of a Y axis, and a direction extending in the up-down direction of the work vehicle **1** is defined as a direction of a Z axis. A+X direction is the right direction, and a -X direction is the left direction. A+Y direction is the front direction, and a -Y direction is the rear direction. A+Z direction is the upper direction, and a -Z direction is the lower direction.

(14) [Wheel Loader]

(15) FIG. 1 is a side view illustrating the work vehicle **1** according to an embodiment. In the embodiment, the work vehicle **1** is a wheel loader. In the embodiment, the work vehicle **1** is referred to as a wheel loader **1** as needed.

(16) As illustrated in FIG. 1, the wheel loader **1** includes a vehicle body **2**, an articulation mechanism **3**, a cab **4**, a travel device **5**, working equipment **6**, and a drive device **7**. The wheel loader **1** travels through the work site by means of the travel device **5**. The wheel loader **1** performs work using the working equipment **6** at the work site.

(17) The vehicle body **2** supports the working equipment **6**. The vehicle body **2** includes a vehicle body front portion **2F** and a vehicle body rear portion **2R**. The vehicle body front portion **2F** is

disposed further on the front side than the vehicle body rear portion 2R. The vehicle body front portion 2F and the vehicle body rear portion 2R are connected by the articulation mechanism 3. The articulation mechanism 3 includes an articulation cylinder 8. The articulation cylinder 8 is a hydraulic cylinder. The articulation cylinder 8 connects the vehicle body front portion 2F to the vehicle body rear portion 2R. When the articulation cylinder 8 extends and retracts, the vehicle body front portion 2F is bent in the right-left direction with respect to the vehicle body rear portion 2R. When the vehicle body front portion 2F is bent with respect to the vehicle body rear portion 2R, the traveling direction of the wheel loader 1 is adjusted.

(18) The cab 4 is supported by the vehicle body 2. In the embodiment, the cab 4 is disposed on the upper portion of the vehicle body rear portion 2R. Inside the cab 4, a seat 9 and an operation lever 10 are disposed. The operator sits on the seat 9. The operation lever 10 is operated by the operator. The operation lever 10 is disposed around the seat 9.

(19) The travel device 5 travels on a ground GR of the work site. The travel device 5 supports the vehicle body 2. The travel device 5 includes a wheel 28. The wheel 28 includes a front wheel 28F attached to the vehicle body front portion 2F and a rear wheel 28R attached to the vehicle body rear portion 2R. The wheel 28 comes into contact with the ground GR. When the wheel 28 rotates, the wheel loader 1 travels.

(20) In the embodiment, the X axis is parallel to a rotation axis CXf of the front wheel 28F. The Z axis is orthogonal to the ground contact surface of the front wheel 28F in contact with the ground GR. When the wheel loader 1 travels in a straight state, the rotation axis CXf of the front wheel 28F and a rotation axis CXr of the rear wheel 28R are parallel to each other.

(21) The working equipment 6 performs work. Examples of the work performed by the working equipment 6 include excavation work, loading work, transportation work, and snow removal work. In the embodiment, the working equipment 6 excavates a target ET. An example of the target ET is a natural ground placed on the ground GR.

(22) FIG. 2 is a diagram illustrating the working equipment 6 and the drive device 7 according to the embodiment. As illustrated in FIGS. 1 and 2, the working equipment 6 is connected to the vehicle body front portion 2F. The working equipment 6 includes a boom 12 and a bucket 13. The working equipment 6 includes a bell crank 14 and a bucket link 15. The drive device 7 operates the working equipment 6. The drive device 7 includes a boom cylinder 18 and a bucket cylinder 19. The boom cylinder 18 is a hydraulic cylinder. The bucket cylinder 19 is a hydraulic cylinder.

(23) The base end portion of the boom 12 is turnably connected to the vehicle body front portion 2F. The boom 12 turns about a turning axis AXa with respect to the vehicle body front portion 2F. A bracket 16 is secured to the middle portion of the boom 12.

(24) The bucket 13 is a work member that excavates the target ET. The bucket 13 holds an excavated object. The bucket 13 has a bottom plate portion 13B. A blade tip portion 13T is provided at the tip portion of the bottom plate portion 13B. The blade tip portion 13T extends in the right-left direction. The bucket 13 has an opening portion 13M.

(25) The base end portion of the bucket 13 is turnably connected to the tip end portion of the boom 12. The bucket 13 turns about a turning axis AXb with respect to the boom 12. The bucket 13 is disposed further on the front side than the front wheel 28F. A bracket 17 is secured to a part of the bucket 13.

(26) The middle portion of the bell crank 14 is turnably connected to the bracket 16 of the boom 12. The bell crank 14 turns about a turning axis AXc with respect to the bracket 16 of the boom 12.

(27) The base end portion of the bucket link 15 is turnably connected to the lower end portion of the bell crank 14. The bucket link 15 turns about a turning axis AXd with respect to the bell crank 14. The tip end portion of the bucket link 15 is turnably connected to the bracket 17 of the bucket 13. The bucket link 15 turns about a turning axis AXe with respect to the bracket 17 of the bucket 13. The bell crank 14 is connected to the bucket 13 via the bucket link 15.

(28) The boom cylinder 18 operates the boom 12. The base end portion of the boom cylinder 18 is

connected to the vehicle body front portion **2F**. The base end portion of the boom cylinder **18** turns about a turning axis **AXf** with respect to the vehicle body front portion **2F**. A bracket **11** is secured to the boom **12**. The tip end portion of the boom cylinder **18** is connected to the bracket **11** of the boom **12**. The boom **12** turns about a turning axis **AXg** with respect to the boom cylinder **18**.
(29) The bucket cylinder **19** operates the bucket **13**. The base end portion of the bucket cylinder **19** is connected to the vehicle body front portion **2F**. The tip end portion of the bucket cylinder **19** is connected to the upper end portion of the bell crank **14**. The bell crank **14** turns about a turning axis **AXh** with respect to the bucket cylinder **19**.

(30) [Operation of Working Equipment]

(31) The working equipment **6** is front-loading-type working equipment in which the opening portion **13M** of the bucket **13** faces to the front side in the excavation work. When the boom cylinder **18** extends and retracts, the boom **12** performs a raising operation or a lowering operation. When the bucket cylinder **19** extends and retracts, the bucket **13** performs a tilt operation or a dumping operation. The excavated object scooped by the blade tip portion **13T** enters the inside of the bucket **13** through the opening portion **13M**.

(32) The raising operation of the boom **12** refers to an operation in which the boom **12** turns about the turning axis **AXa** such that the tip end portion of the boom **12** is away from the ground **GR**. The lowering operation of the boom **12** refers to an operation in which the boom **12** turns about the turning axis **AXa** such that the tip end portion of the boom **12** approaches the ground **GR**.

(33) When the boom cylinder **18** extends, the boom **12** performs the raising operation. When the boom cylinder **18** retracts, the boom **12** performs the lowering operation.

(34) The tilt operation of the bucket **13** refers to an operation in which the bucket **13** turns about the turning axis **AXb** such that the opening portion **13M** of the bucket **13** faces upward and the blade tip portion **13T** is away from the ground **GR**. The dumping operation of the bucket **13** refers to an operation in which the bucket **13** turns about the turning axis **AXb** such that the opening portion **13M** of the bucket **13** faces downward and the blade tip portion **13T** approaches the ground **GR**.

(35) When the bucket cylinder **19** extends, the bell crank **14** turns such that the upper end portion of the bell crank **14** moves to the front side and the lower end portion of the bell crank **14** moves to the rear side. When the lower end portion of the bell crank **14** moves to the rear side, the bucket **13** is pulled to the rear side by the bucket link **15** and performs the tilt operation. When the bucket cylinder **19** retracts, the bell crank **14** turns such that the upper end portion of the bell crank **14** moves to the rear side and the lower end portion of the bell crank **14** moves to the front side. When the lower end portion of the bell crank **14** moves to the front side, the bucket **13** is pushed to the front side by the bucket link **15** and performs the dumping operation.

(36) When the bucket **13** performs the tilt operation, the excavated object is scooped by the bucket **13** and held in the bucket **13**. When the bucket **13** performs the dumping operation, the excavated object held in the bucket **13** is discharged from the bucket **13**.

(37) [Configuration of Work Vehicle]

(38) FIG. **3** is a configuration diagram illustrating the wheel loader **1** according to the embodiment. As illustrated in FIG. **3**, the wheel loader **1** includes a power source **20**, a power take-off (PTO) **21**, the travel device **5**, the drive device **7**, the operation lever **10**, and a control device **50**.

(39) The power source **20** is, for example, a diesel engine. The power source **20** generates power.

(40) The power take-off **21** is connected to the power source **20**. The power take-off **21** distributes power from the power source **20** to the travel device **5** and the drive device **7**. At least part of the power generated in the power source **20** is transmitted to the travel device **5** via the power take-off **21**. At least part of the power generated by the power source **20** is transmitted to the drive device **7** via the power take-off **21**.

(41) The drive device **7** operates the working equipment **6** using the power transmitted from the power source **20** via the power take-off **21**. The drive device **7** includes a hydraulic pump **22**, a boom control valve **23**, a bucket control valve **24**, the boom cylinder **18**, and the bucket cylinder

19.

(42) The hydraulic pump **22** is connected to the power take-off **21**. The hydraulic pump **22** discharges hydraulic oil using the power transmitted from the power source **20**. At least part of the hydraulic oil discharged from the hydraulic pump **22** is supplied to the boom cylinder **18** via the boom control valve **23**. At least part of the hydraulic oil discharged from the hydraulic pump **22** is supplied to the bucket cylinder **19** via the bucket control valve **24**.

(43) The boom control valve **23** controls a direction and a flow rate of hydraulic oil supplied from the hydraulic pump **22** to the boom cylinder **18**. When the direction of the hydraulic oil supplied to the boom cylinder **18** is controlled by the boom control valve **23**, the boom cylinder **18** extends and retracts, and the boom **12** performs the raising operation or the lowering operation. When the flow rate of the hydraulic oil supplied to the boom cylinder **18** is controlled by the boom control valve **23**, the boom cylinder speed indicating the extension/retraction speed of the boom cylinder **18** is controlled, and the movement speed of the boom **12** is controlled. The movement speed of the boom **12** includes the angular velocity of the boom **12** about the turning axis AXa when the boom **12** performs the raising operation and the angular velocity of the boom **12** about the turning axis AXa when the boom **12** performs the lowering operation.

(44) The bucket control valve **24** controls a direction and a flow rate of hydraulic oil supplied from the hydraulic pump **22** to the bucket cylinder **19**. When the direction of the hydraulic oil supplied to the bucket cylinder **19** is controlled by the bucket control valve **24**, the bucket cylinder **19** extends and retracts, and the bucket **13** performs the tilt operation or the dumping operation. When the flow rate of the hydraulic oil supplied to the bucket cylinder **19** is controlled by the bucket control valve **24**, the bucket cylinder speed indicating the extension/retraction speed of the bucket cylinder **19** is controlled, and the movement speed of the bucket **13** is controlled. The movement speed of the bucket **13** includes the angular velocity of the bucket **13** about the turning axis AXb when the bucket **13** performs the tilt operation and the angular velocity of the bucket **13** about the turning axis AXb when the bucket **13** performs the dumping operation.

(45) The travel device **5** travels using the power transmitted from the power source **20** via the power take-off **21**. The travel device **5** includes a clutch **25**, a torque converter **26**, a transmission **27**, and the wheels **28**. The power generated in the power source **20** is transmitted to the wheels **28** via the clutch **25**, the torque converter **26**, and the transmission **27**. The power generated in the power source **20** is transmitted to the wheels **28**, and the wheels **28** rotate, whereby the travel device **5** travels.

(46) The clutch **25** is disposed between the power source **20** and the transmission **27**. The clutch **25** is connected to the power take-off **21**. In the embodiment, the clutch **25** is disposed between the power take-off **21** and the torque converter **26**. The clutch **25** is a wet-type multi-disc hydraulic clutch capable of adjusting the degree of engagement between the input shaft side and the output shaft side. The clutch **25** includes an input side member and an output side member.

(47) The degree of engagement of the clutch **25** changes with the clutch pressure applied to the clutch **25**. The clutch pressure refers to the pressure of the hydraulic oil applied to the input side member and the output side member of the clutch **25**. As the degree of engagement of the clutch **25** is changed, the power transmitted from the power take-off **21** to the torque converter **26** changes.

(48) The degree of engagement of the clutch **25** is adjusted between 0[%] and 100[%]. When the clutch pressure is minimum, the degree of engagement is 0[%]. As the clutch pressure increases, the degree of engagement increases. When the clutch pressure is maximum, the degree of engagement is 100[%]. The state in which the degree of engagement is 0[%] is an open state in which the input side member and the output side member are not engaged and the power from the power take-off **21** to the travel device **5** is not transmitted to the torque converter **26**. The state in which the degree of engagement is 100[%] is a complete engagement state in which the input side member and the output side member are engaged and all the power from the power take-off **21** to the travel device **5** is transmitted to the torque converter **26**. The state in which the degree of

engagement is higher than 0[%] and lower than 100[%] is a partial engagement state in which the input side member and the output side member slip relative to each other and part of the power from the power take-off **21** is thus transmitted to the torque converter **26**. As the degree of engagement of the clutch **25** is adjusted, the transmission rate of the power transmitted from the power take-off **21** to the torque converter **26** is adjusted.

(49) The torque converter **26** is disposed between the clutch **25** and the transmission **27**.

(50) The transmission **27** includes a plurality of speed clutches and a plurality of direction switching clutches. By selectively engaging or disengaging the plurality of speed clutches with or from each other, the speed stage of the transmission **27** is changed. By selectively engaging or disengaging the plurality of direction switching clutches with or from each other, forward movement and backward movement of the travel device **5** are switched.

(51) The operation lever **10** includes a boom operation lever **29**, a bucket operation lever **30**, and a selector lever **31**. When the boom operation lever **29** is operated by the operator, the boom **12** operates. When the bucket operation lever **30** is operated by the operator, the bucket **13** operates. When the selector lever **31** is operated by the operator, the speed stage of the transmission **27** is changed.

(52) When the boom operation lever **29** is operated, a boom operation signal is transmitted from the boom operation lever **29** to the control device **50**. The control device **50** controls the boom control valve **23** on the basis of the boom operation signal from the boom operation lever **29**. When the boom control valve **23** is controlled, the boom cylinder **18** extends and retracts, and the boom **12** is operated. When the bucket operation lever **30** is operated, a bucket operation signal is transmitted from the bucket operation lever **30** to the control device **50**. The control device **50** controls the bucket control valve **24** on the basis of the bucket operation signal from the bucket operation lever **30**. When the bucket control valve **24** is controlled, the bucket cylinder **19** extends and retracts, and the bucket **13** is operated.

(53) When the selector lever **31** is operated, a speed stage operation signal is transmitted from the selector lever **31** to the control device **50**. The control device **50** changes the speed stage of the transmission **27** on the basis of the speed stage operation signal from the selector lever **31**.

(54) The boom operation lever **29** is provided with a kick-down switch **32**. The kick-down switch **32** lowers the speed stage of the transmission **27** in a state where the selector lever **31** is not operated. The low speed stage refers to a speed stage having a large transmission gear ratio.

(55) When the kick-down switch **32** is operated by the operator, a kick-down operation signal is output from the kick-down switch **32** to the control device **50**. Based on the kick-down operation signal from the kick-down switch **32**, the control device **50** changes the speed stage to a lower speed stage than the speed stage of the transmission **27** at the time when the kick-down operation signal is acquired. For example, in a case where the speed stage at the time when the kick-down operation signal is acquired is the second speed stage, the control device **50** changes the speed stage of the transmission **27** from the second speed stage to the first speed stage, which is lower than the second speed stage.

(56) The wheel loader **1** also includes a boom angle sensor **33**, a bucket angle sensor **34**, a pressure sensor **35**, and a vehicle speed sensor **36**.

(57) The boom angle sensor **33** detects a boom angle α indicating the angle of the boom **12**. The boom angle sensor **33** detects the boom angle α with respect to the vehicle body **2** in the local coordinate system. An example of the boom angle sensor **33** is a potentiometer disposed at the connection portion between the vehicle body front portion **2F** and the boom **12**. As illustrated in FIG. 2, the boom angle α is an angle formed by a line **L1** connecting the turning axis **AXa** to the turning axis **AXb** and a line **L2** passing through the turning axis **AXa** and parallel to the line connecting the rotation axis **CXf** to the rotation axis **CXr**. In the embodiment, in a case where the boom **12** is inclined further to the ground **GR** than the line **L2**, the boom angle α is a negative value. When the boom **12** is raised, the boom angle α increases. The detection data of the boom

angle sensor **33** is output to the control device **50**. The control device **50** can calculate a boom angular velocity α_v indicating the angular velocity of the boom **12** about the turning axis AXa on the basis of the detection data of the boom angle sensor **33**. That is, the boom angle sensor **33** can function as an angular velocity sensor that detects the boom angular velocity α_v . Note that the boom angle sensor **33** may be a stroke sensor that detects the stroke of the boom cylinder **18**.

(58) The bucket angle sensor **34** detects a bucket angle β indicating the angle of the bucket **13**. The bucket angle sensor **34** detects a bell crank angle indicating the angle of the bell crank **14** with respect to the boom **12** in the local coordinate system. An example of the bucket angle sensor **34** is a potentiometer disposed at the connection portion between the boom **12** and the bell crank **14**. The bucket angle sensor **34** detects a bell crank angle that is an angle formed by the boom **12** and the bell crank **14**. The bell crank angle and the bucket angle β correspond to each other on a one-to-one basis. As the bell crank angle is detected, the bucket angle β is detected. As illustrated in FIG. 2, the bucket angle β is an angle formed by a line L3 passing through the turning axis AXb and parallel to a bottom surface **132** of the bucket **13** and a line L4 passing through the turning axis AXb and parallel to the line L2. In the embodiment, in a case where the line L3 is inclined further to the ground GR than the line L4, the bucket angle β is a negative value. When the bucket **13** performs the tilt operation, the bucket angle β increases. The detection data of the bucket angle sensor **34** is output to the control device **50**. The control device **50** can calculate a bucket angular velocity β_v indicating the angular velocity of the bucket **13** about the turning axis AXb on the basis of the detection data of the bucket angle sensor **34**. That is, the bucket angle sensor **34** can function as an angular velocity sensor that detects the bucket angular velocity β_v . Note that the bucket angle sensor **34** may be a stroke sensor that detects the stroke of the bucket cylinder **19**.

(59) The pressure sensor **35** detects the pressure of the hydraulic oil in the boom cylinder **18**. The pressure of the hydraulic oil in the boom cylinder **18** includes bottom pressure Pb of the boom cylinder **18**. The detection data of the pressure sensor **35** is output to the control device **50**.

(60) The vehicle speed sensor **36** detects a vehicle speed Vc indicating the traveling speed of the travel device **5**. The detection data of the vehicle speed sensor **36** is output to the control device **50**.

(61) The wheel loader **1** also includes an interface device **37**. The interface device **37** is disposed inside the cab **4**.

(62) The interface device **37** includes an auto-excavation start switch **38**, a sound output unit **39**, and a display unit **40**. The sound output unit **39** outputs, for example, a warning sound. The display unit **40** displays display data related to the state or control of the working equipment **6**.

(63) [Control System]

(64) FIG. 4 is a functional block diagram illustrating a control system **60** of the wheel loader **1** according to the embodiment. The control system **60** includes the control device **50**, the operation lever **10**, the travel device **5**, the drive device **7**, the boom angle sensor **33**, the bucket angle sensor **34**, the pressure sensor **35**, the vehicle speed sensor **36**, and the interface device **37**.

(65) The control device **50** includes a computer system. The control device **50** outputs a control command for controlling the wheel loader **1**.

(66) FIG. 5 is a block diagram illustrating the control device **50** of the wheel loader **1** according to the embodiment. As illustrated in FIG. 5, the control device includes a processor **51**, a main memory **52**, a storage **53**, and an interface **54**. The processor **51** executes a computer program to perform arithmetic processing of the operation of the working equipment **6**. Examples of the processor **51** include a central processing unit (CPU) and a micro processing unit (MPU). Examples of the main memory **52** include a nonvolatile memory and a volatile memory. An example of the nonvolatile memory is a read only memory (ROM). An example of the volatile memory is a random access memory (RAM). The storage **53** is a non-transitory tangible storage medium. Examples of the storage **53** include a magnetic disk, a magneto-optical disk, and a semiconductor memory. The storage **53** may be an internal medium directly connected to the bus of the control device or an external medium connected to the control device **50** via the interface **54** or

a communication line. The storage **53** stores a computer program for controlling the working equipment **6**.

(67) As illustrated in FIG. **4**, the control device **50** individually communicates with the operation lever **10**, the travel device **5**, the drive device **7**, the boom angle sensor **33**, the bucket angle sensor **34**, the pressure sensor **35**, the vehicle speed sensor **36**, and the interface device **37**.

(68) The control device **50** outputs a control command for controlling the clutch **25** to the clutch **25**. The control device **50** outputs a control command for controlling the degree of engagement of the clutch **25** on the basis of the state of the working equipment **6**. The state of the working equipment **6** includes the operation speed of the working equipment **6**. The operation speed of the working equipment **6** includes the bucket angular velocity βv indicating the angular velocity of the bucket **13**. The bucket angular velocity βv is detected by the bucket angle sensor **34**. In the embodiment, the control device **50** outputs a control command for controlling the degree of engagement of the clutch **25** on the basis of the detection data of the bucket angle sensor **34**.

(69) In the embodiment, the control command output from the control device **50** to control the degree of engagement of the clutch **25** is referred to as a clutch command as needed.

(70) The control device **50** outputs a control command for controlling the transmission **27** to the transmission **27**. The control device **50** outputs a control command for changing the speed stage of the transmission **27**. The control device **50** outputs a control command for moving the travel device **5** forward or backward.

(71) In the embodiment, the control command output from the control device **50** to change the speed stage of the transmission **27** is referred to as a speed change command as needed. Also, the control command output from the control device **50** to move the travel device **5** forward is referred to as a forward movement command as needed.

(72) The control device **50** outputs a control command for controlling the working equipment **6** to at least one of the boom control valve **23** and the bucket control valve **24**. The control device **50** outputs a control command to the boom control valve **23** so that the boom **12** performs the raising operation or the lowering operation at a predetermined boom angular velocity αv . The control device **50** outputs a control command to the bucket control valve **24** so that the bucket **13** performs the tilt operation or the dumping operation at a predetermined bucket angular velocity βv .

(73) The control device **50** individually controls the sound output unit **39** and the display unit **40** of the interface device **37**.

(74) [Automatic Excavation Control]

(75) In the embodiment, the wheel loader **1** performs the excavation work of the target ET using the working equipment **6** in a manual operation mode or an auto-excavation mode. In the manual operation, the working equipment **6** operates as the operator operates at least one of the boom operation lever **29** and the bucket operation lever **30**. In the auto-excavation mode, the working equipment **6** operates on the basis of the control command output from the control device **50** without depending on the operation of the boom operation lever **29** and the bucket operation lever **30**. The auto-excavation mode reduces the burden on the operator in the excavation operation.

(76) In the auto-excavation mode, the control device performs auto-excavation control of the working equipment **6**. The control commands output from the control device **50** in the auto-excavation control include an auto-lift command for causing the boom **12** to perform a lift operation and an auto-tilt command for causing the bucket **13** to perform a tilt operation.

(77) In the embodiment, the lift operation of the boom **12** in response to the auto-lift command is referred to as an auto-lift operation as needed, and the tilt operation of the bucket **13** in response to the auto-tilt command is referred to as an auto-tilt operation as needed.

DESCRIPTION OF EMBODIMENTS

(78) FIG. **6(A)** and FIG. **6(B)** are views for explaining an excavation operation of the wheel loader **1** according to the embodiment. As illustrated in FIG. **6(A)**, in the excavation operation, the wheel loader **1** moves forward so as to approach the target ET. When the wheel loader **1** moves forward,

the posture of the working equipment **6** is adjusted so that the bucket **13** excavates the target ET. In the embodiment, the posture of the working equipment **6** is adjusted so that the blade tip portion **13T** of the bucket **13** gets close to or comes into contact with the ground GR. When the wheel loader **1** moves forward in a state where the blade tip portion **13T** is close to or in contact with the ground GR, the blade tip portion **13T** of the bucket **13** is inserted into the lower end portion of the target ET.

(79) To start the auto-excavation control, the operator operates the auto-excavation start switch **38** of the interface device **37**. When the auto-excavation start switch **38** is operated by the operator, an auto-excavation start signal is transmitted from the auto-excavation start switch **38** to the control device **50**. After acquiring the auto-excavation start signal from the auto-excavation start switch **38**, the control device **50** starts the auto-excavation control. The control device **50** causes the display unit **40** to display an indicator indicating that the auto-excavation control is to be performed.

(80) As illustrated in FIG. **6(B)**, in the auto-excavation control, after the blade tip portion **13T** of the bucket **13** is inserted into the target ET, the control device **50** causes the bucket **13** to perform the auto-tilt operation. As a result, the target ET is excavated by the bucket **13**. The bucket **13** scoops the excavated object. The excavated object is held in the bucket **13**.

(81) In the auto-excavation control, in a case where the bucket **13** is inserted into the target ET is caused to perform the auto-tilt operation, the control device **50** starts the auto-tilt operation of the bucket **13** on the basis of the vehicle speed V_c of the travel device **5** and the bottom pressure P_b of the boom cylinder **18**. Also, the control device **50** ends the auto-tilt operation on the basis of the vehicle speed V_c of the travel device **5** and the increase amount of the bottom pressure P_b from the time when the auto-tilt operation is started. That is, in the auto-excavation control, the control device **50** starts the auto-tilt operation of the bucket **13** on the basis of the detection data of the vehicle speed sensor **36** and the detection data of the pressure sensor **35**. The control device **50** ends the auto-tilt operation on the basis of the detection data of the vehicle speed sensor **36** and the increase amount of the detection value of the pressure sensor **35** from the time point when the auto-tilt operation is started.

(82) Note that, in the auto-excavation control, when the operator operates the kick-down switch **32**, the control device **50** changes the speed stage of the transmission **27** to a low speed stage. When the speed stage of the transmission **27** is changed to a low speed stage, the traction force of the wheel loader **1** increases. Since the traction force of the wheel loader **1** increases, the excavation efficiency of the wheel loader **1** is improved.

(83) As described above, in the auto-excavation control, the wheel loader **1** moves forward toward the target ET and inserts the bucket **13** into the target. For example, in a case where the bucket **13** is strongly pressed against the target ET by the forward movement of the wheel loader **1**, a situation in which the wheel loader **1** cannot move may occur. That is, in a case where the bucket **13** is strongly pressed against the target ET, a situation in which the working equipment **6** cannot move or a situation in which the travel device **5** cannot move may occur. The situation in which the working equipment **6** cannot move in the auto-excavation control includes a situation in which the bucket **13** cannot perform the auto-tilt operation. The situation in which the travel device **5** cannot move in the auto-excavation control includes a situation in which the wheel **28** moving forward cannot rotate. When a situation in which the wheel loader **1** cannot move occurs, the work efficiency of the wheel loader **1** is lowered.

(84) In the embodiment, in the auto-excavation control, the control device **50** outputs a control command for controlling the degree of engagement of the clutch **25** on the basis of the state of the working equipment **6**. The state of the working equipment **6** includes the operation speed of the working equipment **6**. The operation speed of the working equipment **6** includes the bucket angular velocity β_v . The bucket angular velocity β_v is detected by the bucket angle sensor **34**. In a case of determining that the bucket **13** and the wheel **28** cannot move on the basis of the detection data of the bucket angle sensor **34**, the control device **50** outputs a control command so that the degree of

engagement of the clutch **25** decreases.

(85) When the wheel loader **1** moves forward toward the target ET in order to excavate the target ET, the degree of engagement of the clutch **25** is set to a high value (for example, 100[%]). In a case where the degree of engagement of the clutch **25** is maintained at a high value even in a situation where the bucket **13** is strongly pressed against the target ET and the bucket **13** and the wheel **28** cannot move, the state in which the bucket **13** is strongly pressed against the target ET is maintained.

(86) In the embodiment, in a case where the wheel loader **1** cannot move in the auto-excavation control, the degree of engagement of the clutch **25** is decreased. As the degree of engagement of the clutch **25** is decreased, the power transmitted from the power take-off **21** to the wheel **28** is decreased. As a result, the force with which the working equipment **6** is pressed against the target ET is decreased. In addition, since the power transmitted from the power take-off **21** to the wheel **28** is decreased, the traction force of the wheel loader **1** is decreased.

(87) The power take-off **21** distributes power from the power source **20** to the travel device **5** and the drive device **7**. As the degree of engagement of the clutch **25** is decreased, the power distributed to the travel device **5** decreases, and the power distributed to the drive device **7** increases. That is, the power for causing the bucket **13** to perform the auto-tilt operation increases. Since the power for causing the bucket **13** to perform the auto-tilt operation increases in a state where the force with which the bucket **13** is pressed against the target ET is decreased, the bucket **13** can perform the auto-tilt operation.

(88) FIG. 7(A) to FIG. 7(H) are timing charts is a timing chart illustrating a control method for the wheel loader **1** according to the embodiment.

(89) A first graph illustrated in FIG. 7(A) illustrates the relationship between the time [sec.] from the start of the auto-excavation control and the vehicle speed V_c [km/h] of the travel device **5**. The vehicle speed V_c is detected by the vehicle speed sensor **36**.

(90) A second graph illustrated in FIG. 7(B) illustrates the relationship between the time [sec.] from the start of the auto-excavation control and the command value [%] of the auto-lift command output from the control device **50**. A state in which the command value of the auto-lift command is 0[%] indicates that the opening degree of the boom control valve **23** for causing the boom **12** to perform the lift operation is in a fully closed state. When the command value of the auto-lift command increases from 0[%], the opening degree of the boom control valve **23** also increases. A state in which the command value of the auto-lift command is 100[%] indicates that the opening degree of the boom control valve **23** for causing the boom **12** to perform the lift operation is in a fully open state. The state in which the command value of the auto-lift command is 0[%] includes a state in which the auto-lift command is not output from the control device **50**. The state in which the command value of the auto-lift command is higher than 0[%] includes a state in which the auto-lift command is output from the control device **50**.

(91) A third graph illustrated in FIG. 7(C) illustrates the relationship between the time [sec.] from the start of the auto-excavation control and the boom angle α [deg]. The boom angle α is detected by the boom angle sensor **33**.

(92) A fourth graph illustrated in FIG. 7(D) illustrates the relationship between the time [sec.] from the start of the auto-excavation control and the bottom pressure P_b [MPa] of the boom cylinder **18**. The bottom pressure P_b is detected by the pressure sensor **35**.

(93) A fifth graph illustrated in FIG. 7(E) illustrates the relationship between the time [sec.] from the start of the auto-excavation control and the command value [%] of the auto-tilt command output from the control device **50**. A state in which the command value of the auto-tilt command is 0[%] indicates that the opening degree of the bucket control valve **24** for causing the bucket **13** to perform the tilt operation is in a fully closed state. When the command value of the auto-tilt command increases from 0[%], the opening degree of the bucket control valve **24** also increases. A state in which the command value of the auto-tilt command is 100[%] indicates that the opening

degree of the bucket control valve **24** for causing the bucket **13** to perform the tilt operation is in a fully open state. The state in which the command value of the auto-tilt command is 0[%] includes a state in which the auto-tilt command is not output from the control device **50**. The state in which the command value of the auto-tilt command is higher than 0[%] includes a state in which the auto-tilt command is output from the control device **50**.

(94) A sixth graph illustrated in FIG. 7(F) illustrates the relationship between the time [sec.] from the start of the auto-excavation control and the bucket angle β . The bucket angle β is detected by the bucket angle sensor **34**.

(95) A seventh graph illustrated in FIG. 7(G) illustrates the relationship between the time [sec.] from the start of the auto-excavation control and the bucket angular velocity β_v . The bucket angular velocity β_v is detected by the bucket angle sensor **34**.

(96) An eighth graph illustrated in FIG. 7(H) illustrates the relationship between the time [sec.] from the start of the auto-excavation control and the command value [%] of the clutch command output from the control device **50** to control the degree of engagement of the clutch **25**. A state in which the command value of the clutch command is 0[%] indicates a state in which the degree of engagement of the clutch **25** is 0%, that is, a state in which the clutch **25** is in an open state. When the command value of the clutch command increases from 0[%], the degree of engagement of the clutch **25** also increases. A state in which the command value of the clutch command is higher than 0[%] and lower than 100[%] indicates a partial engagement state in which the degree of engagement of the clutch **25** is higher than 0[%] and lower than 100 [%]. A state in which the command value of the clutch command is 100[%] indicates a state in which the degree of engagement of the clutch **25** is 100%, that is, a state in which the clutch **25** is in a complete engagement state.

(97) When the auto-excavation start switch **38** is operated by the operator, the auto-excavation control is started at time t_0 . As illustrated in FIG. 6(A), the posture of the working equipment **6** is adjusted so that the bucket **13** gets close to or comes into contact with the ground GR. The wheel loader **1** moves forward at a vehicle speed V_0 toward the target ET.

(98) At time t_1 , the bucket **13** is inserted into the target ET. When the bucket **13** is inserted into the target ET, the vehicle speed V_c decreases and the bottom pressure P_b increases.

(99) At time t_2 after the time t_1 , the auto-lift operation is started. The control device **50** outputs the auto-lift command for causing the boom **12** to perform the auto-lift operation at the time t_2 when a predetermined auto-lift start condition is satisfied. The auto-lift start condition includes a condition that the vehicle speed V_c is lower than a predetermined vehicle speed determination value d and the bottom pressure P_b is higher than a predetermined pressure determination value b . As an example, the vehicle speed determination value d is 3 [km/h], and the pressure determination value b is 5 [MPa]. When the boom **12** performs the lift operation, a load is applied to the wheel **28**, and the slip of the wheel **28** is prevented.

(100) Note that, in the embodiment, the command value of the auto-lift command is decreased at time t_3 after the time t_2 .

(101) At time t_4 after the time t_3 , the auto-tilt operation is started. The control device **50** outputs the auto-tilt command for causing the bucket **13** to perform the auto-tilt operation at the time t_4 when a predetermined auto-tilt start condition is satisfied. The auto-tilt start condition includes a condition that the vehicle speed V_c is lower than a predetermined vehicle speed determination value k and the bottom pressure P_b is higher than a predetermined pressure determination value j . As an example, the vehicle speed determination value k is 1.45 [km/h], and the pressure determination value j is 16 [MPa].

(102) In the embodiment, the control device **50** starts the auto-tilt operation after ending the auto-lift operation. After stopping the output of the auto-lift command, the control device **50** starts the output of the auto-tilt command.

(103) When the bucket **13** performs the auto-tilt operation, the bucket angle β and the bucket

angular velocity βv increase. Also, when the bucket **13** performs the auto-tilt operation, the bottom pressure P_b increases. When the bucket **13** performs the auto-tilt operation, the vehicle speed V_c decreases. A part of the target ET is excavated by the auto-tilt operation of the bucket **13**, and a space is formed in front of the wheel **28**. As a result, the vehicle speed V_c increases.

(104) At time t_5 after the time t_4 , the auto-tilt operation ends. The control device **50** stops the output of the auto-tilt command at the time t_5 when a predetermined auto-tilt end condition is satisfied. The auto-tilt end condition includes a condition that the vehicle speed V_c is higher than a predetermined vehicle speed determination value n and the increase amount of the bottom pressure P_b from the time t_4 when the auto-tilt operation is started is higher than a predetermined pressure determination value m . As an example, the vehicle speed determination value n is 1.3 [km/h], and the pressure determination value m is 3 [MPa].

(105) In the embodiment, the control device **50** starts the auto-lift operation after ending the auto-tilt operation. After stopping the output of the auto-tilt command, the control device **50** starts the output of the auto-lift command.

(106) At time t_6 after the time t_5 , the auto-tilt start condition is satisfied again. At the time t_6 when the auto-tilt start condition is satisfied, the control device **50** stops the output of the auto-lift command and starts the output of the auto-tilt command.

(107) The control device **50** outputs the auto-lift command and the auto-tilt command so that the auto-lift operation and the auto-tilt operation are repeated while outputting the forward movement command to move the travel device **5** forward.

(108) For example, in a case where the bucket **13** is strongly pressed against the target ET by the forward movement of the wheel loader **1**, a situation in which the wheel loader **1** cannot move may occur. That is, in a case where the bucket **13** is strongly pressed against the target ET, the bucket **13** cannot perform the auto-tilt operation, or the travel device **5** cannot move forward. For example, in the period between the time t_6 and time t_7 illustrated in FIG. 7(A) to FIG. 7(H), a situation occurs in which the bucket **13** cannot move, the bucket angular velocity βv decreases, and the vehicle speed V_c approaches 0. Also, a situation occurs in which not only the bucket **13** but also the boom **12** cannot move, the boom angle α does not change, and the bottom pressure P_b reaches relief pressure P_r .

(109) In the embodiment, the control device **50** outputs the clutch command such that the degree of engagement of the clutch **25** decreases when a clutch control start condition in which a state in which the bucket angular velocity βv is lower than a predetermined speed determination value s continues for a specified time t_s or longer is satisfied. As an example, the speed determination value s is 4 [deg/sec.], and the specified time t_s is 0.5 [sec.].

(110) In the example illustrated in FIG. 7(A) to FIG. 7(H), at the time t_7 after the time t_6 , the clutch control for decreasing the degree of engagement of the clutch **25** is started. The control device **50** outputs the clutch command for decreasing the degree of engagement of the clutch **25** at the time t_7 when the clutch control start condition is satisfied.

(111) In the embodiment, the degree of engagement of the clutch **25** is gradually decreased from the time t_7 to time t_8 after the time t_7 . As an example, a decrease rate u of the degree of engagement per unit time is 0.18 [%/sec.]. Also, the degree of engagement of the clutch **25** is controlled so that the degree of engagement does not fall below a predetermined lower limit w . As an example, the lower limit w is 1[%].

(112) As the degree of engagement of the clutch **25** is gradually decreased, the power distributed to the drive device **7** increases, and the bucket **13** starts to move. In the example illustrated in FIG. 7(A) to FIG. 7(H), the bucket **13** starts to move at time t_9 after the time t_8 , and the bucket angular velocity βv gradually increases. The bucket **13** can get out of the situation of being unable to move. Also, a part of the target ET is excavated by the auto-tilt operation of the bucket **13**, and a space is formed in front of the wheel **28**. As a result, the travel device **5** also starts to move, and the vehicle speed V_c starts to increase.

(113) When a clutch control end condition is satisfied, the control device **50** outputs the clutch command for increasing the degree of engagement of the clutch **25**. The clutch control end condition may be a condition that a predetermined time has elapsed from the time **t7** when the clutch control is started. The clutch control end condition may be a condition that a state in which the bucket angular velocity βv is higher than a predetermined speed determination value y continues for a specified time t_y or longer. As an example, the speed determination value y is 1 [deg/sec.], and the specified time t_y is 0.5 [sec.]. Note that the clutch control end condition may be a condition that the bucket angular velocity βv is higher than 0 [deg/sec.]. In the example illustrated in FIG. 7(A) to FIG. 7(H), the clutch control ends at time **t10** after a predetermined time has elapsed from the time **t7**. The time **t10** is later than the time **t9**. At the time **t10**, the bucket angular velocity Bv is higher than the speed determination value s .

(114) In the embodiment, the degree of engagement of the clutch **25** is gradually increased from the time **t10** to time **t11** after the time **t10**. As an example, an increase rate v of the degree of engagement per unit time is 0.18 [%/sec.]. Also, the degree of engagement of the clutch **25** is controlled so that the degree of engagement does not exceed a predetermined upper limit z . As an example, the upper limit z is the degree of engagement before the degree of engagement of the clutch **25** is decreased (the degree of engagement at the time **t7**).

(115) [Control Method for Work Vehicle]

(116) FIGS. **8**, **9**, and **10** are flowcharts illustrating a control method for the wheel loader **1** according to the embodiment.

(117) In the embodiment, the control device **50** defines a plurality of stages indicating the states of the auto-excavation control. The control device **50** defines stages 0 to 7 as stages.

(118) The stage 0 indicates a standby state and an end state of the auto-excavation control. The stage 1 indicates a determination state of a start condition for the auto-excavation control. The stage 2 indicates an execution state of the auto-lift operation. The stage 3 indicates a start standby state of the auto-lift operation. The stage 4 indicates an execution state of the auto-lift operation. The stage 5 indicates an execution state of the auto-tilt operation. The stage 6 indicates a determination state of the auto-tilt end condition and a determination state of the clutch control start condition. The stage 7 indicates an execution state of the auto-lift operation.

(119) The control device **50** determines whether or not the wheel loader **1** is under auto-excavation control (Step **S1**).

(120) In a case where it is determined in Step **S1** that the auto-excavation control is not being performed (Step **S1**: No), the control device **50** determines whether or not the auto-excavation mode is ON. That is, the control device **50** determines whether or not the auto-excavation mode is valid (Step **S2**).

(121) In a case of acquiring the auto-excavation start signal from the auto-excavation start switch **38**, the control device **50** determines that the auto-excavation mode is ON.

(122) In a case where it is determined in Step **S2** that the auto-excavation mode is ON (Step **S2**, Yes), the control device **50** causes the display unit **40** of the interface device **37** to display an indicator indicating that the auto-excavation mode is ON (Step **S3**).

(123) After the indicator is displayed on the display unit **40**, the control device **50** determines whether or not an insertion condition of the wheel loader **1** into the target ET is satisfied (Step **S4**).

(124) The insertion condition is a condition that the wheel loader **1** moves forward and the bucket **13** is close to or in contact with the ground GR. As illustrated in FIG. 6(A), in a case where the auto-excavation control is executed, the posture of the working equipment **6** is adjusted so that the blade tip portion **13T** of the bucket **13** gets close to or comes into contact with the ground GR. The wheel loader **1** moves forward toward the target ET in a state where the posture of the working equipment **6** has been adjusted.

(125) In a case where the forward movement command to the travel device **5** is output, the control device **50** can determine that the wheel loader **1** is moving forward. In a case where the detection

value of the boom angle sensor **33** is lower than a predetermined angle determination value a , the control device **50** can determine that the bucket **13** is close to or in contact with the ground GR. As an example, the angle determination value a is -30 [deg]. As a result, the control device **50** can determine whether or not the insertion condition of the wheel loader **1** is satisfied.

(126) In a case where it is determined in Step **S4** that the insertion condition is satisfied (Step **S4**: Yes), the control device **50** determines whether or not a kick-down condition is satisfied (Step **S5**).

(127) The kick-down condition is a condition that the transmission **27** is set to an automatic speed change mode and a kick-down command is input into the control device or a condition that the speed stage of the transmission **27** is set to the first speed stage and the kick-down switch **32** is operated.

(128) In a case where it is determined in Step **S5** that the kick-down condition is satisfied (Step **S5**: Yes), the control device **50** shifts the state of the auto-excavation control to the stage 1 in order to start the auto-excavation control (Step **S6**).

(129) The control device **50** determines whether or not an end condition of the auto-excavation control is satisfied (Step **S7**).

(130) In the embodiment, the end condition of the auto-excavation control includes at least one of a condition that the auto-excavation mode is OFF, a condition that a different control command from the forward movement command is output to the transmission **27**, a condition that a predetermined time (for example, 0.5 [sec.]) has elapsed since the bucket **13** has reached a tilt end, a condition that the boom angle α is equal to or higher than a predetermined angle (for example, 0 [deg]), a condition that the working equipment **6** is locked, a condition that a failure has occurred in the control system **60** related to the working equipment **6**, a condition that the operation amount of the boom operation lever **29** for causing the boom **12** to perform the lowering operation is larger than a predetermined amount, and a condition that the operation amount of the bucket operation lever **30** for causing the bucket **13** to perform the dumping operation is larger than a predetermined amount.

(131) The tilt end refers to a position in the extension/retraction range of the bucket cylinder **19** where the bucket cylinder **19** extends the most and the bucket **13** cannot perform the tilt operation. The tilt end is detected by the bucket angle sensor **34**.

(132) In a case where it is determined in Step **S7** that the end condition of the auto-excavation control is not satisfied (Step **S7**: No), the control device **50** determines whether or not the state of the auto-excavation control is the stage 1 (Step **S8**).

(133) In a case where it is determined in Step **S8** that the state of the auto-excavation control is the stage 1 (Step **S8**: Yes), the control device **50** determines whether or not an auto-lift start condition is satisfied (Step **S9**).

(134) The auto-lift start condition is a condition for starting the auto-lift operation of the boom **12**. In the embodiment, the auto-lift start condition is a condition that a state in which the vehicle speed V_c of the travel device **5** is lower than the predetermined vehicle speed determination value d continues for a predetermined time t_b or longer, a state in which the bottom pressure P_b of the boom cylinder **18** is higher than the predetermined pressure determination value b continues for a predetermined time t_a or longer, and the boom angle α is lower than a predetermined angle determination value c .

(135) As an example, the vehicle speed determination value d is 3 [km/h], the pressure determination value b is [MPa], the angle determination value c is -10 [deg], the predetermined time t_a is 0.1 [sec.], and the predetermined time t_b is 0.1 [sec.].

(136) In a case where it is determined in Step **S9** that the auto-lift start condition is satisfied (Step **S9**: Yes), the control device **50** shifts the state of the auto-excavation control to the stage 2 in order to start the auto-lift operation (Step **S10**).

(137) The control device **50** determines whether or not the state of the auto-excavation control is the stage 2 (Step **S11**).

(138) In a case where it is determined in Step **S11** that the state of the auto-excavation control is the

stage 2 (Step S11: Yes), the control device 50 determines whether or not an auto-lift end condition is satisfied (Step S12).

(139) The auto-lift end condition is a condition for ending the auto-lift operation of the boom 12. In the embodiment, the auto-lift end condition is a condition that the increase amount of the boom angle α from the time when the boom 12 starts the auto-lift operation is larger than a predetermined angle determination value f or a condition that a state in which the bottom pressure P_b is higher than a predetermined pressure determination value g continues for a predetermined time t_c or longer.

(140) As an example, the angle determination value f is 2 [deg], the pressure determination value g is 30 [MPa], and the predetermined time t_c is 0.1 [sec.].

(141) In a case where it is determined in Step S12 that the auto-lift end condition is satisfied (Step S12: Yes), the control device 50 shifts the state of the auto-excavation control to the stage 3 in order to stand by for the start of the auto-lift operation (Step S13).

(142) The control device 50 initializes a counter t_m of clutch control of the clutch 25 (Step S14). That is, the control device 50 sets the counter t_m to 0.

(143) The control device 50 determines whether or not the state of the auto-excavation control is the stage 3 (Step S15).

(144) In a case where it is determined in Step S15 that the state of the auto-excavation control is the stage 3 (Step S15: Yes), the control device 50 sets the command value of the auto-tilt command to 0[%] and sets the command value of the auto-lift command to 0[%] (Step S16).

(145) FIG. 11 is a diagram for describing an auto-tilt cycle table according to the embodiment. The auto-tilt command is output on the basis of the auto-tilt cycle table. The auto-tilt cycle table is started when the stage 3 is started. As illustrated in FIG. 11, the auto-tilt cycle table includes an ON time Δt_1 for opening the bucket control valve 24 to cause the bucket 13 to perform the tilt operation and an OFF time Δt_2 for closing the bucket control valve 24. In the auto-tilt cycle table, a plurality of ON times Δt_1 and a plurality of OFF times Δt_2 are set on the basis of the number of times of the auto-tilt operation scheduled to be executed. The auto-tilt cycle table is stored in advance in the control device 50.

(146) In the embodiment, at least four ON times Δt_1 and four OFF times Δt_2 are repeated. Note that FIG. 11 schematically illustrates two ON times Δt_1 and one OFF time Δt_2 . The first ON time Δt_1 is set to 0 [sec.]. The first OFF time Δt_2 is set to 0.4 [sec.]. The second ON time Δt_1 is set to 0.6 [sec.]. The second OFF time Δt_2 is set to [sec.]. The third ON time Δt_1 is set to 0.6 [sec.]. The third OFF time Δt_2 is set to 0.3 [sec.]. The fourth ON time Δt_1 is set to 0.6 [sec.]. The fourth OFF time Δt_2 is set to 0.2 [sec.]. Note that the fifth and subsequent ON times Δt_1 are set to 0.6 [sec.], and the fifth and subsequent OFF times Δt_2 are set to 0.2 [sec.].

(147) The control device 50 determines whether or not the OFF time Δt_2 has elapsed since the auto lift end condition is satisfied (Step S17).

(148) In the embodiment, the control device 50 does not start the auto-tilt operation until the OFF time Δt_2 elapses after the previous auto-tilt end condition of the boom 12 is satisfied.

(149) In a case where it is determined in Step S17 that the OFF time Δt_2 has elapsed (Step S17: Yes), the control device 50 shifts the state of the auto-excavation control to the stage 4 in order to execute the auto-lift operation (Step S18).

(150) The control device 50 determines whether or not the state of the auto-excavation control is the stage 4 (Step S19).

(151) In a case where it is determined in Step S19 that the state of the auto-excavation control is the stage 4 (Step S19: Yes), the control device 50 determines whether or not the boom angle α is lower than a predetermined angle determination value r (Step S20).

(152) As an example, the angle determination value r is -33 [deg].

(153) In a case where it is determined in Step S20 that the boom angle α is lower than the angle determination value r (Step S20: Yes), the control device 50 sets the command value of the auto-tilt

command to 0, and sets the command value of the auto-lift command to a command value q , which is higher than 0 (Step S21).

(154) As an example, the command value q is 80[%].

(155) Subsequently, the control device **50** determines whether or not the auto-tilt start condition is satisfied (Step S22).

(156) The auto-tilt start condition is a condition for starting the auto-tilt operation of the bucket **13**. The auto-tilt start condition is the same as the auto-lift end condition. In the embodiment, the auto-tilt start condition is a condition that a state in which the vehicle speed V_c is lower than the predetermined vehicle speed determination value k continues for a predetermined time t_d or longer, and a state in which the bottom pressure P_b is higher than the predetermined pressure determination value j continues for a predetermined time t_e or longer.

(157) As an example, the vehicle speed determination value k is 1.45 [km/h], the pressure determination value j is 16 [MPa], the predetermined time t_d is 0.1 [sec.], and the predetermined time t_e is 0.1 [sec.].

(158) In a case where it is determined in Step S22 that the auto-tilt start condition is satisfied (Step S22: Yes), the control device **50** shifts the state of the auto-excavation control to the stage 5 in order to execute the auto-tilt operation (Step S23).

(159) The control device **50** determines whether or not the state of the auto-excavation control is the stage 5 (Step S24).

(160) In a case where it is determined in Step S24 that the state of the auto-excavation control is the stage 5 (Step S24: Yes), the control device **50** sets the command value of the auto-lift command to 0 and sets the command value of the auto-tilt command to a command value p , which is higher than 0 (Step S25).

(161) As an example, the command value p is 80[%]. When the auto-tilt command is output, the bucket **13** starts the auto-tilt operation.

(162) The control device **50** determines whether or not the ON time Δt_1 has elapsed since the auto-tilt start condition is satisfied (Step S26).

(163) In a case where it is determined in Step S26 that the ON time Δt_1 has elapsed (Step S26: Yes), the control device **50** shifts the state of the auto-excavation control to the stage 6 in order to determine the satisfaction of the auto-tilt end condition and determine the satisfaction of the clutch control start condition (Step S27).

(164) The control device **50** determines whether or not the state of the auto-excavation control is the stage 6 (Step S28).

(165) In a case where it is determined in Step S28 that the state of the auto-excavation control is the stage 6 (Step S28: Yes), the control device **50** determines whether or not the clutch control start condition is satisfied (Step S29).

(166) The clutch control start condition is a condition for starting the decrease of the degree of engagement of the clutch **25**. In the embodiment, the clutch control start condition is a condition that a state in which the bucket angular velocity β_v is lower than the predetermined speed determination value s is continued for the specified time t_s or longer.

(167) As an example, the speed determination value s is 4 [deg/sec.], and the specified time t_s is 0.5 [sec.].

(168) In a case where it is determined in Step S29 that the clutch control start condition is satisfied (Step S29: Yes), the control device **50** sets the counter t_m of clutch control of the clutch **25** to a predetermined value t_n (Step S30).

(169) As an example, the predetermined value t_n is 50.

(170) The control device **50** determines whether or not the auto-tilt end condition is satisfied (Step S31).

(171) The auto-tilt end condition is a condition for ending the auto-tilt operation. In the embodiment, the auto-tilt end condition is a condition that the bottom pressure P_b is higher than the

pressure determination value j and the increase amount of the bottom pressure P_b from the time when the bucket **13** starts the auto-tilt operation is larger than the predetermined pressure determination value m or a condition that a state in which the vehicle speed V_c is higher than the vehicle speed determination value n continues for a predetermined time t_f or longer.

(172) As an example, the vehicle speed determination value n is 1.3 [km/h], the pressure determination value m is 3 [MPa], and the predetermined time t_f is 0.1 [sec].

(173) In a case where it is determined in Step **S31** that the auto-tilt end condition is satisfied (Step **S31**: Yes), the control device **50** sets the command value of the auto-tilt command to 0[%]. Also, the control device **50** adds 1 to the current number of times of the auto-tilt operation (Step **S32**).

(174) When the auto-tilt command is set to 0[%], the operation of the bucket cylinder **19** stops, and the auto-tilt operation of the bucket **13** ends.

(175) In order to stand by for the start of the auto-lift operation, the control device **50** shifts the state of the auto-excavation control to the stage 3 (Step **S33**).

(176) The control device **50** determines whether or not the state of the auto-excavation control is the stage 7 (Step **S34**).

(177) In a case where it is determined in Step **S34** that the state of the auto-excavation control is the stage 7 (Step **S34**: Yes), the control device **50** determines whether or not the boom angle α is lower than the angle determination value r (Step **S35**).

(178) In a case where it is determined in Step **S35** that the boom angle α is lower than the angle determination value r (Step **S35**: Yes), the control device **50** sets the command value of the auto-lift command to a command value x (Step **S36**).

(179) As an example, the command value x is 80[%].

(180) The control device **50** determines whether or not the clutch control of the clutch **25** is continued (Step **S37**).

(181) In the embodiment, the control device **50** determines that the clutch control is continued in a case where the counter t_m is higher than 0.

(182) In a case where it is determined in Step **S37** that the control of the clutch **25** is continued (Step **S37**: Yes), the control device **50** decreases the command value of the clutch command by the decrease rate u [%/sec] (Step **S38**). The decrease rate u indicates a decrease rate of the degree of engagement of the clutch **25** per unit time. As an example, the decrease rate u is 0.18 [%/sec]. Also, the clutch command (the degree of engagement of the clutch **25**) is controlled so that the value does not fall below the predetermined lower limit w . As an example, the lower limit w is 1[%].

(183) The control device **50** subtracts 1 from the counter t_m (Step **S39**).

(184) The control device **50** determines whether or not the state of the auto-excavation control is the stage 0 (Step **S40**).

(185) In a case where it is determined in Step **S40** that the state of the auto-excavation control is not the stage 0 (Step **S40**: No), the control device **50** sets a boom command to a value obtained by adding the current auto-lift command to a current lever command (Step **S41**).

(186) In addition, the control device **50** sets a bucket command to a value obtained by adding the current auto-tilt command to the current lever command (Step **S42**).

(187) The lever command is a control command for determining the opening degree of the boom control valve **23** or the opening degree of the bucket control valve **24** obtained from the operation amount of the boom operation lever **29** or the bucket operation lever **30**. In a case where the respective operation levers (**29** and **30**) are not operated, the lever command becomes 0.

(188) The control device **50** determines whether or not a completion condition of the auto-excavation control is satisfied (Step **S43**).

(189) The completion condition of the auto-excavation control is a condition that the excavation by means of the auto-excavation control is completed. In the embodiment, the completion condition of the auto-excavation control is a condition that a predetermined time t_g has elapsed after the bucket **13** reaches the tilt end in the auto-excavation control. As an example, the predetermined time t_g is

0.5 [sec.].

(190) In a case where it is determined in Step S43 that the completion condition of the auto-excavation control is satisfied (Step S43: Yes), the control device 50 causes the sound output unit 39 of the interface device 37 to output a completion sound indicating that the excavation by means of the auto-excavation control is completed (Step S44).

(191) The sound is output from the sound output unit 39 to enable the operator of the wheel loader 1 to know that the excavation by means of the auto-excavation control has been completed.

(192) In order to start the auto-lift operation after the detection of the tilt end, the control device 50 shifts the state of the auto-excavation control to the stage 7 (Step S45).

(193) The control device 50 determines whether or not to end the auto-excavation control (Step S46).

(194) For example, in a case where the operator of the wheel loader 1 cancels the auto-excavation mode, in a case where the operator operates the boom operation lever 29 to cause the boom 12 to perform the lowering operation, or in a case where the operator operates the bucket operation lever 30 to cause the bucket 13 to perform the dumping operation, the operation of the operator is prioritized, and the auto-excavation control by means of the control device 50 ends.

(195) In a case where it is determined in Step S46 that the auto-excavation control is not to be ended (Step S46: No), the control device 50 returns to the start and executes the processing of Step S1 and subsequent steps.

(196) In a case where it is determined in Step S46 that the auto-excavation control is to be ended (Step S46: Yes), the control device 50 ends the auto-excavation control, and causes the sound output unit 39 of the interface device 37 to output an incompleteness sound indicating that the auto-excavation control is ended in an incomplete manner (Step S47).

(197) The incompleteness sound is output from the sound output unit 39 to enable the operator of the wheel loader 1 to know that the auto-excavation control has been ended halfway.

(198) The completion sound output in Step S44 is made different from the incompleteness sound output in Step S47. By doing so, the operator can distinguish the case where the excavation by means of the auto-excavation control has been completed from the case where the auto-excavation control has been ended in an incomplete manner. After the end of Step S47, the control device 50 returns to the start and executes the processing of Step S1 and subsequent steps.

(199) In a case where it is determined in Step S1 that the auto-excavation control is being performed (Step S1: Yes), the control device 50 executes the processing of Step S7 and subsequent steps.

(200) In a case where it is determined in Step S2 that the auto-excavation mode is OFF (Step S2: No), the control device 50 erases the indicator indicating that the auto-excavation mode is ON from the display unit 40 of the interface device 37 (Step S48).

(201) After executing the processing of Step S48, the control device 50 shifts the state of the auto-excavation control to the stage 0 in order to end the auto-excavation control (Step S49).

(202) After the processing of Step S49 is ended, the control device 50 executes the processing of Step S34 and subsequent steps.

(203) In a case where it is determined in Step S4 that the insertion condition is not satisfied (Step S4: No), the control device 50 does not execute the auto-excavation control and executes the processing in Step S34 and subsequent steps.

(204) In a case where it is determined in Step S5 that the kick-down condition is not satisfied (Step S5: No), the control device 50 does not execute the auto-excavation control and executes the processing in Step S34 and subsequent steps.

(205) In a case where it is determined in Step S7 that the end condition of the auto-excavation control is satisfied (Step S7: Yes), the control device 50 does not execute the auto-excavation control and executes the processing in Step S34 and subsequent steps.

(206) In a case where it is determined in Step S8 that the state of the auto-excavation control is not

the stage 1 (Step S8: No), the control device 50 executes the processing of Step S11 and subsequent steps.

(207) In a case where it is determined in Step S9 that the auto-lift start condition is not satisfied (Step S9: No), the control device 50 does not start the auto-lift operation and executes the processing of Step S34 and subsequent steps.

(208) In a case where it is determined in Step S11 that the state of the auto-excavation control is not the stage 2 (Step S11: No), the control device 50 executes the processing of Step S15 and subsequent steps.

(209) In a case where it is determined in Step S12 that the auto-lift end condition is not satisfied (Step S12: No), the control device 50 does not end the auto-lift operation and determines whether or not a neutral state of the boom operation lever 29 has continued for a predetermined time t_h (Step S50).

(210) As an example, the predetermined time t_h is 0.1 [sec.].

(211) In a case where it is determined in Step S50 that the neutral state of the boom operation lever 29 has continued for the predetermined time t_h (Step S50: Yes), the control device 50 sets the command value of the auto-lift command to a command value h (Step S51).

(212) As an example, the command value h is 80[%].

(213) In a case where it is determined in Step S50 that the neutral state of the boom operation lever 29 has not continued for the predetermined time t_h (Step S50: No), it means that the operator of the wheel loader 1 has operated the boom operation lever 29. The control device 50 sets the auto-lift command to 0[%] in order to prioritize the operation of the operator (Step S52).

(214) After executing the processing of Step S51 or Step S52, the control device 50 executes the processing of Step S34 and subsequent steps.

(215) In a case where it is determined in Step S15 that the state of the auto-excavation control is not the stage 3 (Step S15: No), the control device 50 executes the processing of Step S19 and subsequent steps.

(216) In a case where it is determined in Step S17 that the OFF time Δt_2 has not elapsed (Step S17: No), the control device 50 executes the processing of Step S19 and subsequent steps.

(217) In a case where it is determined in Step S19 that the state of the auto-excavation control is not the stage 4 (Step S19: No), the control device 50 executes the processing of Step S24 and subsequent steps.

(218) In a case where it is determined in Step S20 that the boom angle α is equal to or higher than the angle determination value r (Step S20: No), the control device 50 executes the processing of Step S24 and subsequent steps.

(219) In a case where it is determined in Step S22 that the auto-tilt start condition is not satisfied (Step S22: No), the control device 50 executes the processing of Step S24 and subsequent steps.

(220) In a case where it is determined in Step S24 that the state of the auto-excavation control is not the stage 5 (Step S24: No), the control device 50 executes the processing of Step S28 and subsequent steps.

(221) In a case where it is determined in Step S26 that the ON time Δt_1 has not elapsed (Step S26: No), the control device 50 executes the processing of Step S28 and subsequent steps.

(222) In a case where it is determined in Step S28 that the state of the auto-excavation control is not the stage 6 (Step S28: No), the control device 50 executes the processing of Step S34 and subsequent steps.

(223) In a case where it is determined in Step S29 that the clutch control condition is not satisfied (Step S29: Yes), the control device 50 does not execute the processing of Step S30 and executes the processing of Step S31 and subsequent steps.

(224) In a case where it is determined in Step S34 that the state of the auto-excavation control is not the stage 7 (Step S34: No), the control device 50 executes the processing of Step S37 and subsequent steps.

(225) In a case where it is determined in Step S35 that the boom angle α is equal to or higher than the angle determination value r (Step S35: No), the control device 50 executes the processing of Step S37 and subsequent steps.

(226) In a case where it is determined in Step S37 that the clutch control of the clutch 25 is not continued (Step S37: No), the control device 50 increases the command value of the clutch command by the increase rate v [%/sec] (Step S53). The increase rate v indicates an increase rate of the degree of engagement of the clutch 25 per unit time. As an example, the increase rate v is 0.18 [%/sec.]. Also, the command value of the clutch command (the degree of engagement of the clutch 25) is controlled so that the value does not exceed the predetermined upper limit z . As an example, the upper limit z is the degree of engagement before the degree of engagement of the clutch 25 is decreased. After executing the processing of Step S53, the control device 50 executes the processing of Step S40 and subsequent steps.

(227) In a case where it is determined in Step S40 that the state of the auto-excavation control is the stage 0 (Step S40: Yes), the control device 50 executes the processing of Step S46 and subsequent steps.

(228) In a case where it is determined in Step S43 that the completion condition of the auto-excavation control is not satisfied (Step S43: No), the control device 50 executes the processing of Step S46 and subsequent steps.

(229) [Effects]

(230) As described above, in the embodiment, the degree of engagement of the clutch 25 is controlled on the basis of the state of the working equipment 6. As the degree of engagement of the clutch 25 is increased, the traction force of the wheel loader 1 (the grip force of the wheel 28 with respect to the ground GR) is increased. Thus, the wheel loader 1 can excavate the target ET. In a case where the working equipment 6 and the wheel 28 are strongly pressed against the target ET by the forward movement of the wheel loader 1, and a situation in which the wheel loader 1 cannot move occurs, the degree of engagement of the clutch 25 is decreased, whereby the wheel loader 1 gets out of a state where the traction force has reached the maximum. By doing so, the bucket 13 can perform the tilt operation. In this manner, by decreasing the degree of engagement of the clutch 25, the wheel loader 1 can get out of a situation of being unable to move in excavation of the target ET.

(231) In the embodiment, the control device 50 controls the degree of engagement of the clutch 25 on the basis of the operation speed of the working equipment 6. When the working equipment 6 and the wheel 28 are strongly pressed against the target ET by the forward movement of the wheel loader 1, the working equipment 6 cannot move. In a case where it is determined that the working equipment 6 cannot move on the basis of the operation speed of the working equipment 6, the degree of engagement of the clutch 25 is decreased, whereby the working equipment 6 can get out of a situation of being unable to move.

(232) When the clutch control end condition is satisfied, the control device 50 outputs the clutch command for decreasing the degree of engagement of the clutch 25. In the embodiment, the clutch control start condition is a condition that a state in which the operation speed of the working equipment 6 is lower than the speed determination value s continues for the specified time is or longer. By defining the clutch control start condition, the control device 50 can appropriately decrease the degree of engagement of the clutch 25.

(233) The power generated by the power source 20 is distributed to the travel device 5 and the drive device 7 by the power take-off 21. When a situation in which the wheel loader 1 cannot move occurs and the degree of engagement of the clutch 25 is decreased, the power distributed to the drive device 7 that drives the working equipment 6 increases. As a result, the working equipment 6 and the wheel 28 can get out of the situation of being unable to move.

(234) In the excavation operation, the bucket 13 of the working equipment 6 is inserted into the target ET. The control device 50 can appropriately control the degree of engagement of the clutch

25 so that the bucket **13** that cannot move can move on the basis of the bucket angular velocity β_v .
(235) The control device **50** starts the tilt operation of the bucket **13** when the auto-tilt start condition defined on the basis of the vehicle speed V_c of the travel device **5** and the bottom pressure P_b of the boom cylinder **18** is satisfied. Also, the control device **50** ends the tilt operation of the bucket **13** when the auto-tilt end condition defined on the basis of the vehicle speed V_s of the travel device **5** and the increase amount of the bottom pressure P_b from the time when the tilt operation is started is satisfied. By doing so, the tilt operation of the bucket **13** is started and ended at the appropriate time.

(236) [Other Embodiments]

(237) In the above-described embodiment, the control device controls the degree of engagement of the clutch **25** on the basis of the bucket angular velocity β_v as the operation speed of the working equipment **6**. The control device **50** may control the degree of engagement of the clutch **25** on the basis of the boom angular velocity α_v as the operation speed of the working equipment **6**. The control device **50** may control the degree of engagement of the clutch **25** in a case of determining that the boom **12** cannot move on the basis of the boom angular velocity α_v .

(238) In the above-described embodiment, the control device **50** controls the degree of engagement of the clutch **25** on the basis of the operation speed of the working equipment **6** including at least one of the bucket angular velocity β_v and the boom angular velocity α_v as the state of the working equipment **6**. The control device **50** may control the degree of engagement of the clutch **25** on the basis of the pressure of the hydraulic oil in the boom cylinder **18** as the state of the working equipment **6**. As described above, when the boom **12** cannot move, a situation occurs in which the bottom pressure P_b of the boom cylinder **18** reaches the relief pressure P_r . The control device **50** may decrease the degree of engagement of the clutch **25** in a case of determining that the pressure of the hydraulic oil in the boom cylinder **18** has reached the relief pressure. Also, the control device **50** may decrease the degree of engagement of the clutch **25** in a case of determining that the pressure of the hydraulic oil in the bucket cylinder **19** has reached the relief pressure.

(239) In the above-described embodiment, the auto-excavation start switch **38** and the kick-down switch **32** may be used at the same time. By using the auto-excavation start switch **38** and the kick-down switch **32** at the same time, the speed stage of the transmission **27** is changed to a low speed stage at the same time as the start of the auto-excavation mode. Thus, the excavation operation is easily and efficiently performed.

REFERENCE SIGNS LIST

(240) **1** WHEEL LOADER (WORK VEHICLE) **2** VEHICLE BODY **2F** VEHICLE BODY FRONT PORTION **2R** VEHICLE BODY REAR PORTION **3** ARTICULATION MECHANISM **4** CAB **5** TRAVEL DEVICE **6** WORKING EQUIPMENT **7** DRIVE DEVICE **8** ARTICULATION CYLINDER **9** SEAT **10** OPERATION LEVER **11** BRACKET **12** BOOM **13** BUCKET **13B** BOTTOM PLATE PORTION **13M** OPENING PORTION **13T** BLADE TIP PORTION **14** BELL CRANK **15** BUCKET LINK **16** BRACKET **17** BRACKET **18** BOOM CYLINDER **19** BUCKET CYLINDER **20** POWER SOURCE **21** POWER TAKE-OFF **22** HYDRAULIC PUMP **23** BOOM CONTROL VALVE **24** BUCKET CONTROL VALVE **25** CLUTCH **26** TORQUE CONVERTER **27** TRANSMISSION **28** WHEEL **28F** FRONT WHEEL **28R** REAR WHEEL **29** BOOM OPERATION LEVER **30** BUCKET OPERATION LEVER **31** SELECTOR LEVER **32** KICK-DOWN SWITCH **33** BOOM ANGLE SENSOR **34** BUCKET ANGLE SENSOR **35** PRESSURE SENSOR **36** VEHICLE SPEED SENSOR **37** INTERFACE DEVICE **38** AUTO-EXCAVATION START SWITCH **39** SOUND OUTPUT UNIT **40** DISPLAY UNIT **50** CONTROL DEVICE **51** PROCESSOR **52** MAIN MEMORY **53** STORAGE **54** INTERFACE **60** CONTROL SYSTEM **132** BOTTOM SURFACE AX_a TURNING AXIS AX_b TURNING AXIS AX_c TURNING AXIS AX_d TURNING AXIS AX_e TURNING AXIS AX_f TURNING AXIS AX_g TURNING AXIS AX_h TURNING AXIS CX_f ROTATION AXIS CX_r ROTATION AXIS ET TARGET GR GROUND $L1$ LINE $L2$ LINE $L3$ LINE $L4$ LINE P_b BOTTOM PRESSURE V_c VEHICLE SPEED α BOOM

Claims

1. A work vehicle comprising: a power source; a travel device that includes a clutch capable of adjusting a degree of engagement between the power source and a transmission, and travels on a basis of power transmitted from the power source; working equipment; a drive device that operates the working equipment on a basis of power transmitted from the power source; and a control device that outputs a control command for controlling the degree of engagement of the clutch on a basis of a state of the working equipment, wherein the state of the working equipment includes an operation speed of the working equipment, and wherein the control device outputs the control command such that the degree of engagement decreases when a clutch control start condition is satisfied in which a state in which the operation speed of the working equipment is lower than a speed determination value continues for a specified time or longer.
 2. The work vehicle according to claim 1, comprising a power take-off that distributes power from the power source to the travel device and the drive device, wherein, as the degree of engagement is decreased, power distributed to the drive device increases.
 3. The work vehicle according to claim 1, wherein the working equipment includes a boom and a bucket, and wherein the operation speed of the working equipment includes an angular velocity of the bucket.
 4. The work vehicle according to claim 3, wherein the drive device includes a boom cylinder that operates the boom and a bucket cylinder that operates the bucket, and wherein the control device starts a tilt operation of the bucket on a basis of a traveling speed of the travel device and a pressure of hydraulic oil in the boom cylinder, and ends the tilt operation on a basis of the traveling speed of the travel device and an increase amount of the pressure from a time when the tilt operation is started.
 5. A control method for a work vehicle comprising: in the work vehicle including working equipment, outputting a control command for performing engagement of a clutch disposed between a power source and a transmission and capable of adjusting a degree of engagement such that power generated in the power source is transmitted to a travel device including the clutch; and outputting a control command for decreasing the degree of engagement of the clutch on a basis of a state of the working equipment, wherein the state of the working equipment includes an operation speed of the working equipment, and wherein the control command is output such that the degree of engagement decreases when a clutch control start condition is satisfied in which a state in which the operation speed of the working equipment is lower than a speed determination value continues for a specified time or longer.
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